Informatica® Cloud Application Integration
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Amazon Bedrock Chat with History

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Table of Contents

Preface 5

Chapter 1: Introduction to Amazon Bedrock Chat with History recipe 6

Amazon Bedrock Chat with History recipe contents 6

Amazon Bedrock Chat with History recipe assets 7

Amazon Bedrock Chat with History process 7

Chapter 2: Using the Amazon Bedrock Chat with History recipe 8

Step 1. Copy and access the recipe 8

Step 2. Configure and publish the FileConnectionChatHistory connection 9

Step 3. Configure and publish the AmazonBedrockConnection connection 9

Step 4. Configure and publish the processes 9

Step 5. Invoke the process 10

Preface

Use *Amazon Bedrock Chat with History* to learn how to maintain chat history in a file and use it as context for the next user query and the Large Language Model's (LLM) response. This guide assumes that you have an understanding of the Amazon Bedrock Connector concepts.

CHAPTER 1

Introduction to Amazon Bedrock Chat with History recipe

The Amazon Bedrock Chat with History recipe is initiated using an HTTP request. You can maintain the chat history in a file and use it as context for the next user query and the Large Language Model's (LLM) response.

The process maintains chat history in a file and uses it as context for the last user question and LLM answer. It reads input parameters to check if the file exists, retrieves the last user question and LLM answer, prepares a request, and sends it. If the file doesn't exist, it creates one and writes the user input and LLM answer to it.

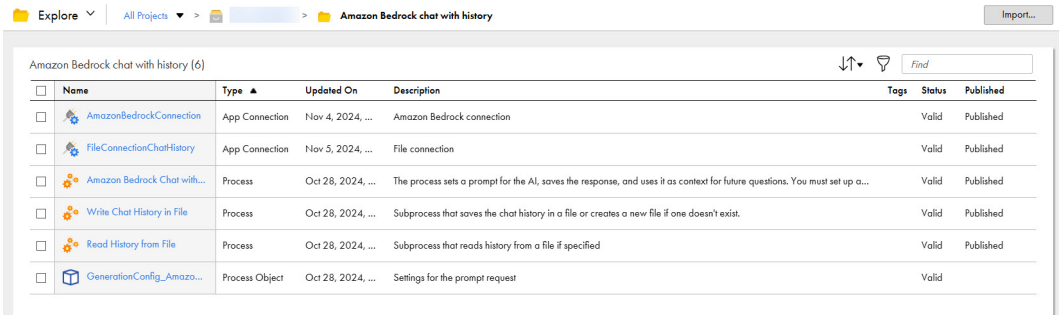
You can use the following file formats:

`.txt`, `.xml`, `.json`, `.html`, `.csv`, `.md`, and `.yaml`

Amazon Bedrock Chat with History recipe contents

The Amazon Bedrock Chat with History recipe contains a process object, two app connections, and three processes.

The following image shows the assets that the Amazon Bedrock Chat with History recipe package contains:



Amazon Bedrock Chat with History recipe assets

The following table lists the assets that the Amazon Bedrock Chat with History recipe package contains:

Asset Name	Asset Type	Description
AmazonBedrockConnection	App connection	Amazon Bedrock connection.
FileConnectionChatHistory	App connection	File connection.
GenerationConfig_AmazonBedrock_PO	Process object	Creates the prompt request.
Read History from File	Process	The subprocess reads history from a file if specified.
Write Chat History in File	Process	The subprocess saves the chat history in a file or creates a new file if one doesn't exist.
Amazon Bedrock Chat with History	Process	The process sets a prompt, saves the response, and uses it as context for future questions. You must set up a Secure Agent to store and retrieve the history.

Amazon Bedrock Chat with History process

The Amazon Bedrock Chat with History process is called by an HTTP request. It maintains chat history in a file and uses it as context for the last user question and LLM answer.

The process is called by an HTTP request using the following input parameters configured on the **Input Fields** tab of the **Start** Step:

- User_Prompt
- Path_To_File_With_History
- User_ID

The User_ID is used as the file name.

After configuring the prompt instructions, the process reads the file. If no file exists or the file is empty, the process prepares a request without history, and then creates or updates the file. It writes the last question and its LLM response in the following format:

```
Question_to_LLM: user question
LLM_Answer: Amazon Bedrock answer
```

If the file contains history, the process uses only the last question and response. The process then updates the file and appends the new question and answer. You can include additional parsing in a specific location and also include new changes to the request.

CHAPTER 2

Using the Amazon Bedrock Chat with History recipe

To use the Amazon Bedrock Chat with History recipe, you must perform the following steps manually:

1. Copy and access the recipe
2. Configure and publish the FileConnectionChatHistory connection
3. Configure and publish the AmazonBedrockConnection connection
4. Configure and publish the processes
5. Invoke the process

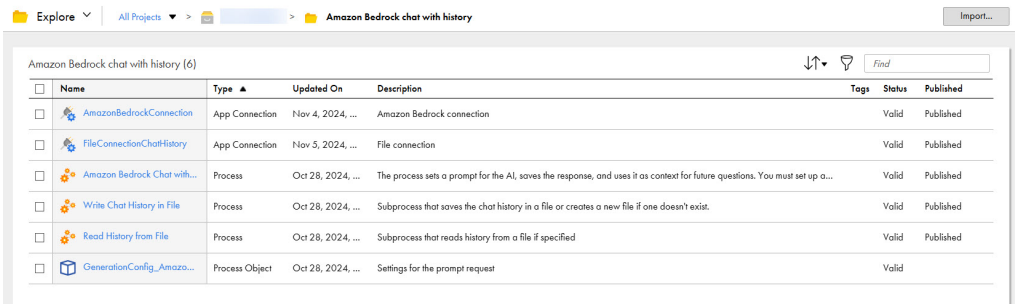
Step 1. Copy and access the recipe

Copy the pre-configured assets in the recipe to a separate project or folder.

1. Open the **Amazon Bedrock Chat with History** recipe and click **Use**.
2. Select the location where you want to copy the recipe, and then click **Continue**.
3. In the **Copying the recipe** dialog box, click **OK**.

   It might take some time for the recipe to get copied. You will receive a notification when the recipe is ready for use.
4. After the recipe is copied, click **Explore** to access the recipe content.
5. Navigate to the project or folder where you copied the recipe or enter the recipe name in the **Find** box.

   All the assets in the recipe are displayed as shown in the following image:



Step 2. Configure and publish the FileConnectionChatHistory connection

Configure the file directory and specify the action to be taken if a file already exists, and then publish the FileConnectionChatHistory connection.

1. Open the **FileConnectionChatHistory** connection.
2. From the **Type** list, select **File**.
3. From the **Runtime Environment** list, select the Secure Agent.
4. On the **Event Targets** tab, in the **Directory** field, enter the directory to store the files. In the **File Exists** field, select **Append**.
5. Save and publish the connection.

Step 3. Configure and publish the AmazonBedrockConnection connection

Configure the access key, secret key, and region to connect to Amazon Bedrock, and then publish the AmazonBedrockConnection connection.

1. Open the **AmazonBedrockConnection** connection.
2. In the **Connection Properties** section, enter values for the following properties:

Property	Description
AccessKey	The access key to access Amazon Bedrock.
SecretKey	The secret key to access Amazon Bedrock.
Region	The region that hosts Amazon Bedrock.

3. Save and publish the connection.

Step 4. Configure and publish the processes

Configure the deployment details of the LLM model and publish the processes.

1. Open the **Read History from File** process.
2. On the **Start** tab of the **Start** step, select the same Secure Agent that you had selected in the **Run On** field for the **FileConnectionChatHistory** connection.
3. Save and publish the process.
4. Open the **Write Chat History in File** process.
5. Optionally, in the **Prepare History to Save in File** step, click the **Assignments** tab. Open the Expression Editor for the **File_Name** field and enter the format to save the file. For example, `.txt`

6. Save and publish the process.
7. Open the **Amazon Bedrock Chat with History** parent process.
8. On the **Temp Fields** tab of the **Start** step, in the **Model_id** field, enter the user-specific deployment ID of the LLM model.

   The default model id is **anthropic.claude-3-sonnet-20240229-v1:0**. You can optionally edit the model id.
9. Optionally, in the **Configure Request Parameters** step, click the **Assignments** tab. Open the Expression Editor for the **Prompt_Configuration** field and enter the prompt instructions as shown in the following sample code:

```
<GenerationConfig_AwsBedrock_PO-2>
   <Temperature>0.5</Temperature>
   <TopP>0.6</TopP>
   <MaxTokens>500</MaxTokens>
</GenerationConfig_AwsBedrock_PO-2>
```

   For the **Prompt_Configuration** field, enter values for the following properties:

Property	Description
maxTokens	Defines the maximum number of tokens that the model can generate in its response. Setting a limit ensures that the response is concise and fits within the desired length constraints.
temperature	Controls the randomness of the model's output. A lower value makes the output more deterministic, while a higher value increases randomness and creativity. For example, a temperature of 0.9 balances between deterministic and creative outputs.
topP	An alternative to sampling with temperature where the model considers the results of the token with topP probability. For example, if topP is set to 0.1, the model considers only the top 10% most probable tokens at each step.

10. Save and publish the process.

Step 5. Invoke the process

The Amazon Bedrock Chat with History process is called by an HTTP request. When you invoke the process, it creates a file with history after your first question.

1. Open the **Amazon Bedrock Chat with History** process and click **Actions** > **Properties Detail** > **Copy Service URL**.
2. Open a text editor and add the input fields and values to the service URL as shown in the following format:

```
<Cloud Application Integration POD URL>/active-bpel/public/rt/<API_name>?
User_Prompt=<User_Prompt>&Path_To_File_With_History=<C:/folder>/&User_ID=<User_ID>
```

3. Open a browser and paste the service URL.

You can also invoke the process using the Run Using option, REST or SOAP API endpoints in any API clients such as cURL, Postman, SOAP UI, or through any programming language.